DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

HERIOT-WATT UNIVERSITY

F78AA Actuarial and Financial Mathematics A

Date: 17 December 2014

Time: 09.00 – 11.00

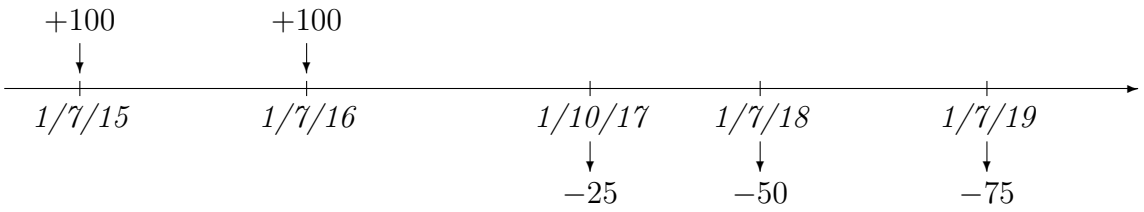
Question	Marks
1	7
2	6
3	10
4	10
5	7
6	10
7	10
Total Marks	60

Attempt ALL questions.

Approved electronic calculators may be used.

This version created on: November 12, 2014

1. The effective rate of interest will be 5% per annum during 2015, 2.5% per quarter-year during 2016 and 2017, and 1% per month thereafter. A series of cashflows is given as follows:



(a) What is the present value on 1 January 2015 of these cashflows?

[5 marks]

(b) What is the accumulation on 1 January 2021 of these cashflows?

[2 marks]

(Minor discrepancies caused by different roundings will not be penalised.)

[Total 7 marks]

2. An annuity-certain is payable annually in advance for 15 years. The first payment is £1,000, and subsequent payments increase at rate 0.04 (that is, by 4% compound) each year. During the first 5 years the effective rate of interest is 7% per annum, thereafter it is 10% per annum. What is the present value of this annuity at outset?

[6 marks]

3. (a) You borrow £200,000 to buy a house. The loan is to be repaid over 30 years by level annual repayments, payable in arrear. The interest rate is 8% per annum effective.

(i) Show that the level annual repayment is £17,765.49.

(ii) Show that the capital outstanding just after the 10th repayment has been made is £174,424.20.

(iii) What is the interest content of the 11th repayment?

[4 marks]

(b) Just after the 10th repayment has been made, you ask the lender if you can miss the next three repayments. The lender agrees, but will charge a rate of interest of 10% per annum effective for the next *six* years, thereafter reverting to 8% per annum effective. What is the new level annual repayment?

[6 marks]

(Minor discrepancies caused by different roundings will not be penalised.)

[Total 10 marks]

continued ...

4. A company issues a bond with term 20 years, to be redeemed at a premium of 2%, with a coupon of 10% per annum payable half-yearly in arrear. Some of the bond is purchased at outset by an investor who pays income tax at 25% and capital gains tax at 15%.

(a) The investor pays a price P per £100 nominal, such that their net redemption yield is 9% per annum effective. What is P?

[5 marks]

(b) Immediately after payment of the 20th coupon, the same investor sells the bond for exactly the same price P per £100 nominal as they paid for it. What net yield per annum does the investor achieve on the completed transaction?

[4 marks]

(c) If, in part (b), the sale had taken place half-way between the payment of the 19th and 20th coupons, for the same price P, would the investor have achieved a higher yield, a lower yield or the same yield?

[1 mark]

[Total 10 marks]

5. An investment fund has assets worth £100,000 on 1 January 2014. It receives an inflow of £20,000 on 1 January 2015 and pays an outflow of £50,000 on 1 January 2017. On 1 January 2018 it has assets of £90,000. There are no other cashflows into or out of the fund during this period.

(a) Suppose that the fund has assets of £130,000 on 1 January 2016. Find the linked internal rate of return during the calendar years 2014–2017, based on the subdivision of the period into the calendar years 2014–2015 and 2016–2017.

[5 marks]

(b) Suppose also that the fund has assets of £102,000 on 1 January 2015 and £129,000 on 1 January 2017, just before any cash inflows or outflows on those days. Find the time weighted rate of return during the calendar years 2014–2017.

[2 marks]

[Total 7 marks]

continued ...

6. (a) State the Principle of Equivalence, and explain how it may be used as a criterion for deciding whether or not to buy an investment.

[2 marks]

(b) You decide to buy whichever of three investments A, B or C has the highest yield. All have the same price now, £20,000.

- Investment A will pay a lump sum of £50,000 at the end of 10 years.
- Investment B will pay £2,200 at the end of each of the next 10 years, plus a lump sum of £20,000 at the end of the 10 years.
- Investment C will pay £3,600 at the end of each of the next 10 years.

Calculate the yield on each investment.

[8 marks]

[Total 10 marks]

7. An investor considers buying an apartment block and renting out the apartments. The price for the apartment block is £6 million. The investor expects to receive £25,000 of rental income at the end of each month for the next five years. Furthermore, the investor expects to sell the apartment block for £7 million after the five years.

The amount of her own money the investor has available to purchase the apartment block is £5 million. The remaining £1 million can be borrowed at a rate of interest of 0.5% per month effective. The investor can lend money at a rate of interest of 0.4% per month effective.

(a) Find the discounted payback period (DPP) for the £1 million loan if the investor is using all income associated with this apartment block to repay the loan as soon as possible. [5 marks]

(b) Find the accumulated profit at the end of the five years of the investment in the apartment block assuming that the investor is using all income associated with this investment to repay the loan as soon as possible. [Hint: How should you allow for the £5 million of the investor's own money that is invested in the project?] [5 marks]

[Total 10 marks]

Total 60 marks